

CIS241

System-Level Programming and Utilities

SSH

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

First, Linux

What is it?

- Often referred to as an operating system (OS)
 - Competing with Mac/Windows
- *Technically*, it is a kernel

Many OSs use the Linux kernel

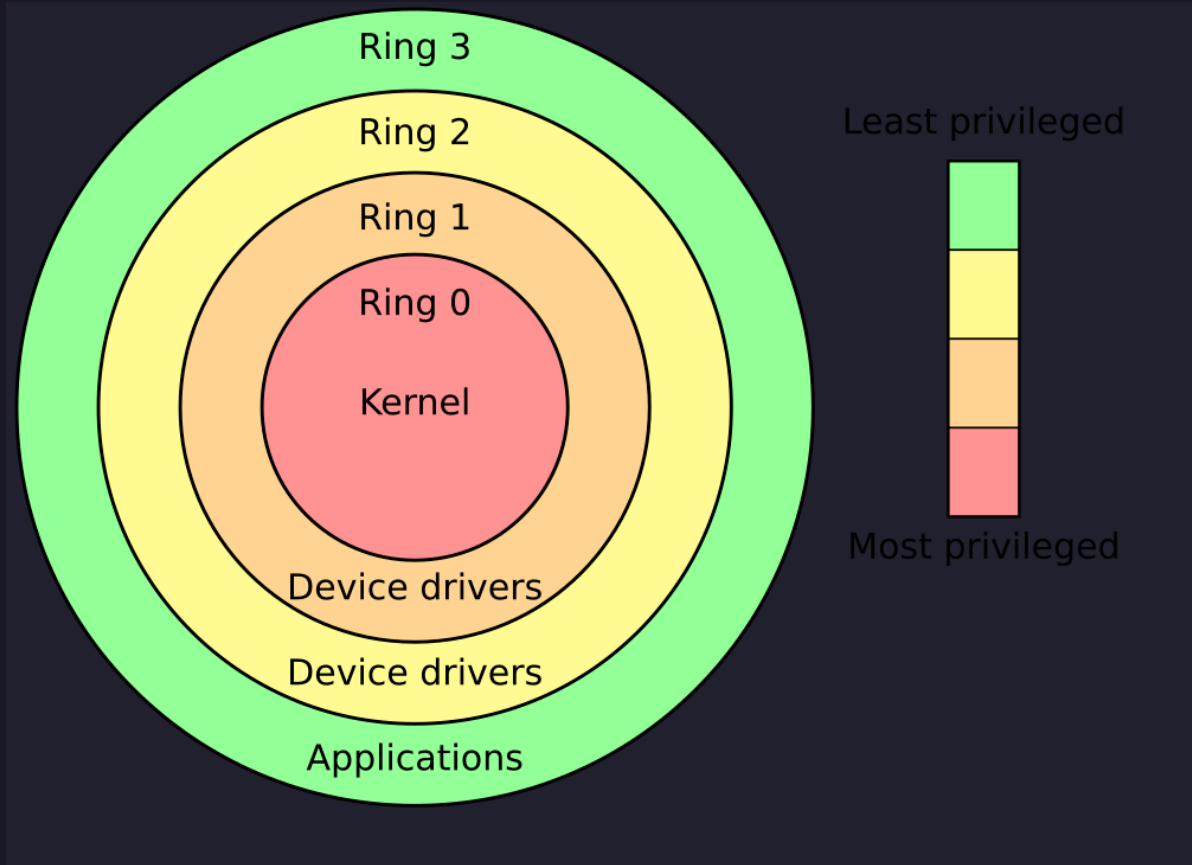
- Ubuntu (what is on EOS)
- Arch, Kali, Gentoo, Fedora, etc.

Isn't Mac Linux too?

- Nope, Darwin! (Unix-based)



Protection Rings



A bit of history

1970s: UNIX source code typically stripped from distribution

- Exorbitant amounts of money to acquire source

Linux first developed by Linus Torvalds to be a copy of UNIX

- Released it to Internet for suggestions on how to improve it
- Kernel only updated by core developers and from suggestions
- Short history of Linux



Unix?

Unix was an earlier operating system

- Linux and other systems are heavily influenced by Unix

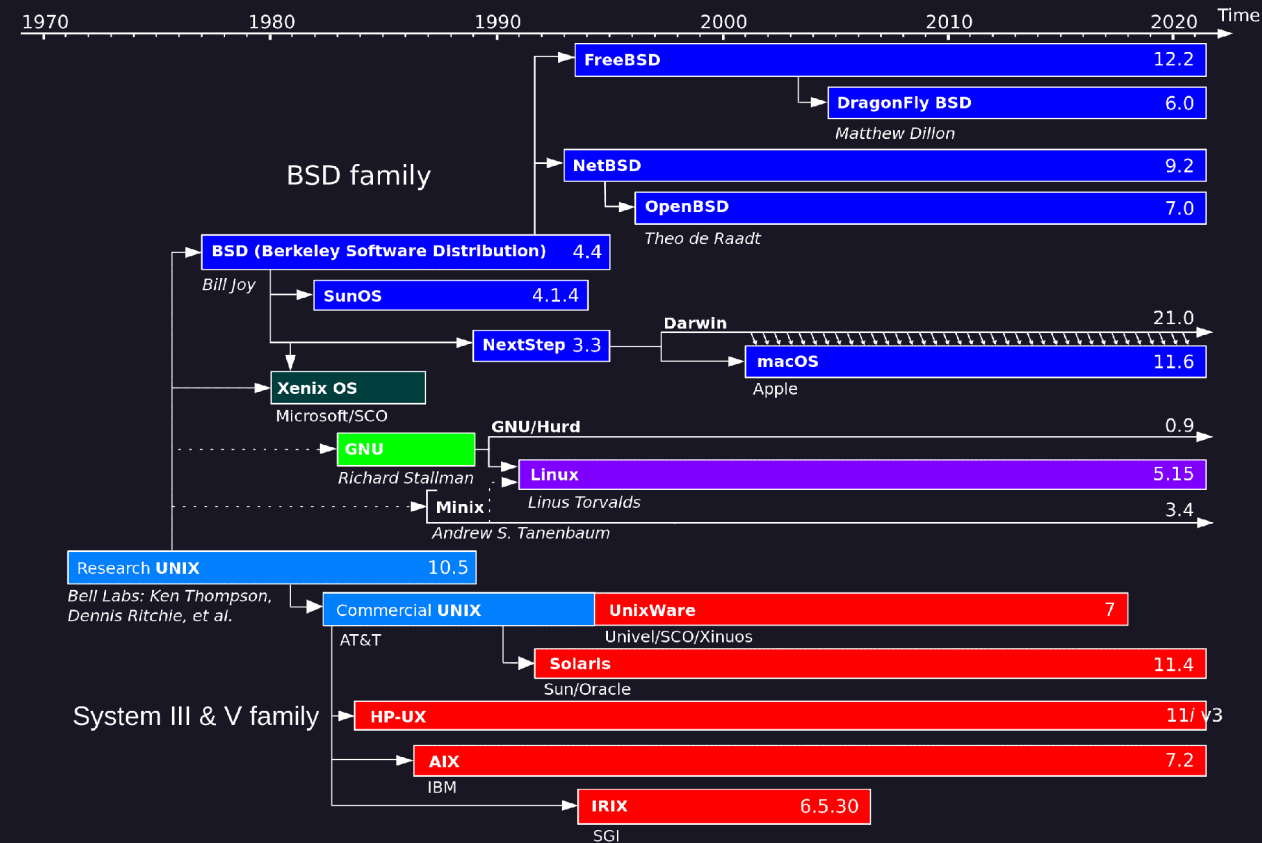
We now use “Unix-like” to describe systems

- They follow the same general ideas as Unix
- This means most things (commands, properties) carry over!

"Unix-like" systems:

- Linux
- macOS
- FreeBSD
- Solaris

Unix Timeline



Advantages of a Linux system

- Risk reduction
- Business needs
- Stability and security
- Flexibility
- Customization
- Support
- Cost reduction

Advantages - Risk reduction

- Changes in the market or customer needs may cause companies to change software frequently
- Can be costly and time-consuming
- Support for closed source software may end
- Vendor may go out of business
- Software version may be retired
- OSS products offer the opportunity to maintain and change the source code

Advantages - Business needs

- Common software available for Linux includes:
- Scientific and engineering software
- Software emulators
- Web servers, web browsers, and e-commerce suites
- Desktop productivity software
- Graphics manipulation software
- Database software
- Security software

Advantages - Stability and security

- Customers using a closed source OS must rely on the OS vendor to fix any bugs
 - Waiting for a hot fix may take weeks or months
- The collaborative open source approach to testing and fixing bugs increases the stability of Linux
- Bugs and security loopholes in OSS programs can be identified and fixed quickly
 - Code is freely available and scrutinized by many developers

Advantages - Flexibility

- Partial list of hardware platforms on which Linux can run:
 - Linux can be (and is) customized to work on mobile and embedded devices

Intel x86/x64	M68K
Itanium	PA-RISC
Mainframe (S/390)	SPARC
ARM	Ultra-SPARC
Alpha	PowerPC
MIPS	

Advantage - Customization

Ability to control the inner workings of the OS

- To use Linux as an Internet Web server, recompile the kernel to include only the support needed to be an Internet Web server
 - Results in a much smaller and faster kernel
- Can choose to install only software packages needed to perform required tasks
- Linux supports several programming languages, such as shell and PERL scripts to customize or automate tasks

Advantage - Support

Linux documentation can be easily found on the Internet

- Frequently asked questions (FAQs)
- HOWTO documents

HOWTO documents are maintained by their authors but are centrally collected by the Linux Documentation Project (LDP)

Linux newsgroups

Linux User Groups (LUGs): Open forum of Linux users who discuss and assist each other in using and modifying the Linux OS

Advantage - Cost reduction

Linux is less expensive than other OSs

- There is no cost associated with acquiring the software
- A wealth of OSS can run on a variety of different hardware platforms running Linux

The largest costs associated with Linux:

- Costs associated with hiring people to maintain the Linux system
- Potential support from vendors (i.e., Red Hat)

Total cost of ownership (TCO): overall cost of using a particular OS

Disadvantages of a Linux system

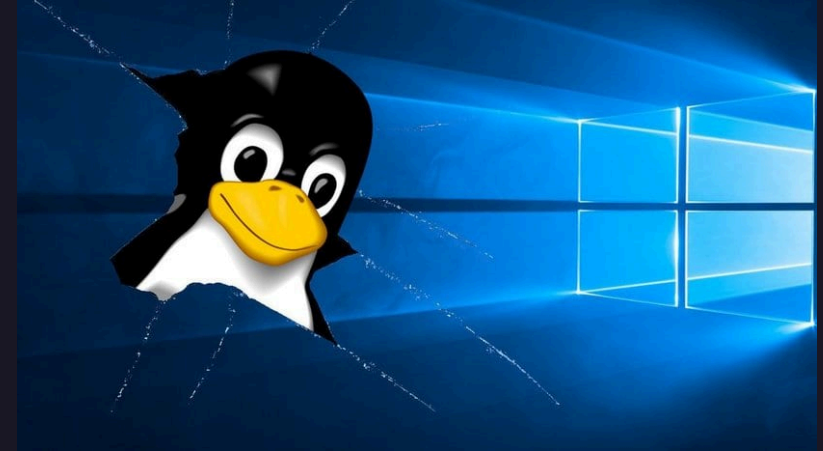
No standard edition of Linux

Support means that you'll be searching for very specific fixes

- Browsing SuperUser/ServerFault, forums, etc.

Not as easy to use as closed-source (Windows)

Program support



Common uses of Linux

- Internet servers
- File/print server
- Application server
- Cloud servers
- Supercomputing
- Workstations (Office, personal, scientific, etc.)
- Mobile



Windows

Windows (today), is not Unix-like

- Can you do your homework on a Windows machine?
- Not by default

However, we have WSL! (🐧_🐧🌸)

- Windows Subsystem for Linux
- If you are on Windows, you must install WSL
<https://learn.microsoft.com/en-us/windows/wsl/install>
 - Be very happy about this - I dreamed about just having the ability to SSH from the terminal when in industry

Terminal

A screenshot of a terminal window on a Linux system. The left side shows a large ASCII art graphic of a mountain range made of the character '7'. The right side displays system information for a Pop!_OS 22.04 LTS system, including host name, kernel, uptime, packages, shell, display resolution, desktop environment, window manager, theme, icons, font, cursor, terminal, CPU, GPU, memory, swap, disk usage, local IP, battery status, and locale. The bottom left shows the terminal prompt 'erik@worktop:~\$' and the bottom right shows a color calibration bar.

Shells (local)

See the little dollar sign?

Shell!

What shell are we using?

```
echo $0
```

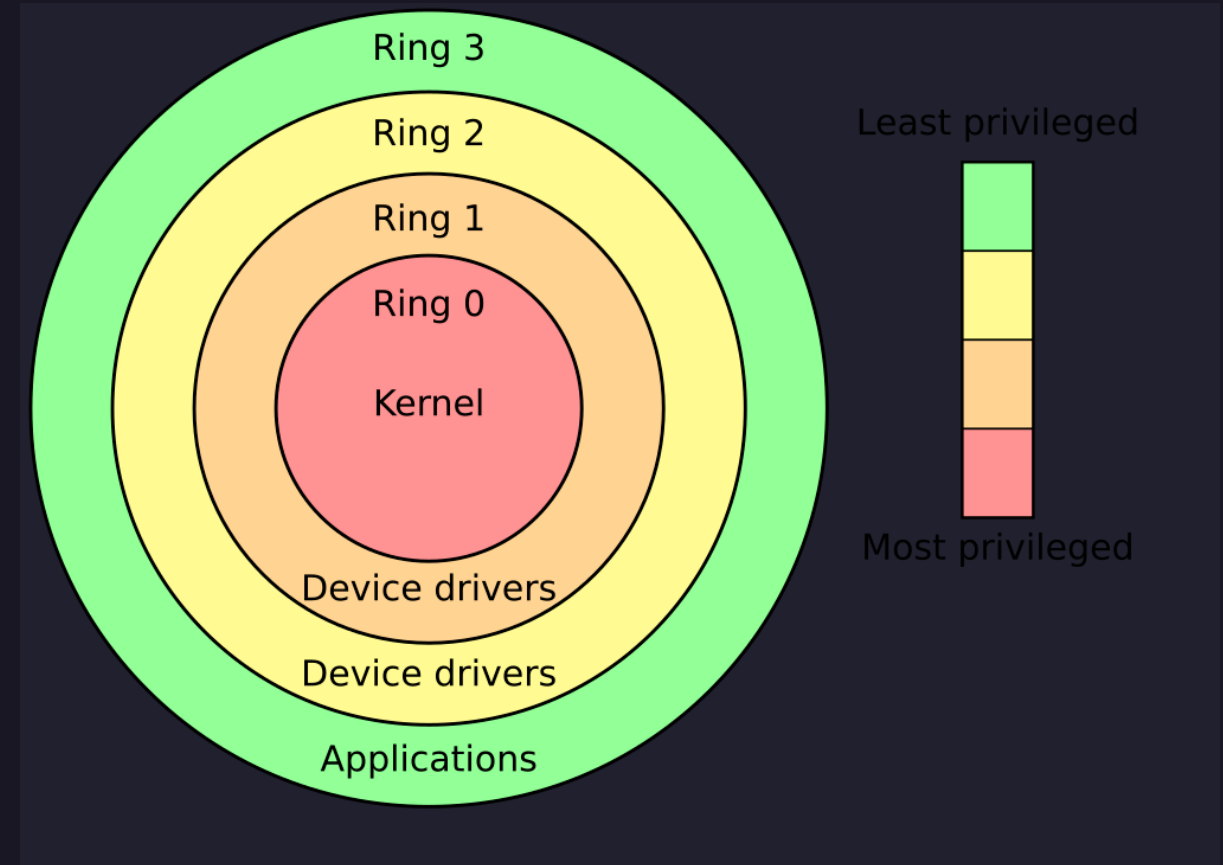
```
erik@worktop: ~  
  
/////////////////  
/////////////////  
767/////////////////  
767676*/////////////////  
/7676767/////////////////  
//*76767/////////////////  
///76767.///7676*/////////  
6//76767///767676/////////  
7676767///76767/////////  
676676///76767/////////  
676,////////767/////////  
7676////////76/////////  
/7676////////767/////////  
//7676///767/////////  
/////////'/////////  
767676767676767,/////////  
6767676767676767/////////  
/////////  
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/////////  
  
erik@worktop  
-----  
OS: Pop!_OS 22.04 LTS x86_64  
Host: Latitude 7400  
Kernel: Linux 6.12.10-76061203-generc  
Uptime: 2 days, 34 mins  
Packages: 4267 (dpkg), 7 (flatpak-us)  
Shell: bash 5.1.16  
Display (AU0243D): 1920x1080 @ 60 Hz]  
DE: GNOME 42.9  
WM: Mutter (X11)  
WM Theme: Pop-dark  
Theme: Pop-dark [GTK2/3/4]  
Icons: Pop [GTK2/3/4]  
Font: Fira Sans (10pt, SemiLight) [G]  
Cursor: Pop (24px)  
Terminal: GNOME Terminal 3.44.0  
Terminal Font: Fira Mono (12pt)  
CPU: Intel(R) Core(TM) i7-8665U (8) z  
GPU: Intel UHD Graphics 620 @ 1.15 G]  
Memory: 15.70 GiB / 31.16 GiB (50%)  
Swap: 0 B / 20.00 GiB (0%)  
Disk (/): 421.50 GiB / 928.94 GiB (44  
Disk (/recovery): 2.84 GiB / 3.99 GiB  
Local IP (wlo1): 192.168.1.46/24  
Battery (DELL 5VC2M85): 100% [AC Con]  
Locale: en_US.UTF-8  
  
$ █
```

Shells (local)

Where is the shell here?

- Ring 3, typically!

How do we run in other rings?



You can use `eos01.cis.gvsu.edu` - `eos30.cis.gvsu.edu` !

Shells (remote)

Note - if you're off-campus you'll need a VPN!

Or, connecting to another machine

- Or or, **doing your homework** ټ_ټ

`ssh` --> Secure Shell

Usage: `ssh username@address`

For example, connecting to EOS:

`ssh username@eos1.cis.gvsu.edu`

Lab fun

DO NOT

SHUT DOWN

THESE MACHINES



Last - the web interface

<https://computing.gvsu.edu>

This will provide a web-based point of access for you!

- **Allows for desktop or terminal**
- **Note that the user experience is not great!**
 - Can't copy/paste files from your computer to the desktop
 - Random connection breaks common
- **However - don't need a VPN!**

Ok now really the last

To exit an SSH session (or to kill your terminal)

```
exit
```